Programme	:	B.Tech., Integrated M.Tech	**Semester**	:	Fall Semester 2026-27
Course	:	Computer Networks	**Code**	:	CSE3006
Faculty	:	Dr.S.Periyanayagi	**Slot/ Class No.**	:	B11+B12+B13, BL2026270100338
Tutorial No.	:	3	**Max. Marks**	:	50 Marks
Date of Submission: 05.09.2026					

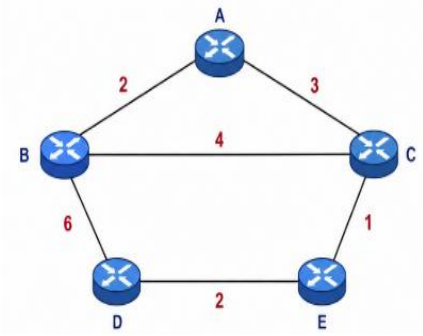
1. In a TCP/IP protocol layer. If the application program creates a message of 200 bytes and each layer (including the fifth and the first) adds a header of 10 bytes to the data unit, what is the efficiency (the ratio of application layer bytes to the number of bytes transmitted) of the system?
2. A network using CSMA/CD has a bandwidth of 5 Mbps. If the maximum propagation time including the delays in the devices and ignoring the time needed to send a jamming signal is 25.6 μs. Calculate the minimum size of the frame.
3. Rewrite the following IP addresses using binary notation:
   a. 110.11.5.88
   b. 12.74.16.18
   c. 201.24.44.32
4. Rewrite the following IP addresses using dotted-decimal notation:
   a. 01011110 10110000 01110101 00010101
   b. 10001001 10001110 11010000 00110001
   c. 01010111 10000100 00110111 00001111
5. An ISP is providing an IP address 192.100.0.30./24 for ABC company which has 3 departments. The details are;
   Product Development - Department1: 60 users (subnet 1- 192.100.0.30 /25)
   Training & Testing - Department2: 50 users (subnet 2 -192.100.0.30 /26)
   HR - Department3: 30 users (subnet 3 -192.100.0.30 /27)
   Analyse and create 3 subnets (one for each department) and assign IP addresses for each user of the subnet. Draw the figure showing the configuration and address range for each department of ABC company.
6. Combine the following three blocks of addresses into a single block:
   a. 16.27.24.0/26
   b. 16.27.24.64/26
   c. 16.27.24.128/25
7. An organization is granted the IP address block 172.20.0.0/16. The network administrator wants to divide this block into 1024 equal-sized subnets.
   a. Find the number of addresses available in each subnet.
   b. Determine the new subnet prefix and corresponding subnet mask.
   c. Find the first and last address of the first subnet.
   d. Find the first and last address of the last subnet.
8. During an online examination, thousands of students simultaneously access the university server to download questions and upload their answers. The sudden increase in traffic causes some routers to become overloaded, resulting in long delays, packet loss, and buffer overflow.

As a network administrator, explain how the following mechanisms can be used to control the congestion:

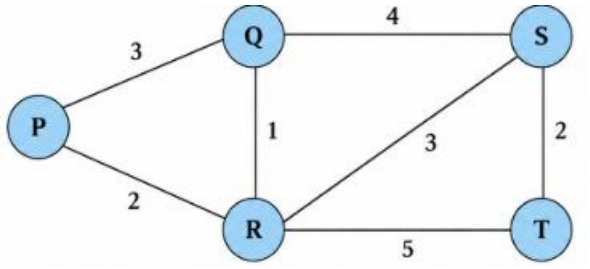
(i) Backpressure
(ii) Choke Packet
(iii) Implicit Signaling
(iv) Explicit Signaling

For each mechanism, explain how congestion is detected and what action is taken to reduce the congestion.

9. A university network provides Internet access to 500 students. During peak hours, a large number of packets arrive at the router simultaneously, causing traffic bursts, congestion, and packet loss. The network administrator decides to use the Leaky Bucket algorithm to regulate the outgoing traffic.
   The bucket can hold 10 packets, and packets are transmitted at a constant rate of 2 packets/second. At a particular instant, 8 packets arrive, followed by 6 packets after 1 second.
   a. Explain how the Leaky Bucket algorithm handles this traffic.
   b. Determine whether any packets will be discarded due to bucket overflow.
   c. Calculate the number of packets transmitted during the given period.
   d. Explain how Leaky Bucket improves QoS.
10. Link State Algorithm:



   a. Explain how router A obtains information about the network topology using the Link State approach.
   b. Construct the Link State Database for router A based on the information available in the network.
   c. Determine the shortest path from router A to every other router using Dijkstra's Shortest Path First algorithm. Show the intermediate steps.
   d. Construct the shortest-path tree rooted at router A and give the final routing table containing the destination, shortest path, and total cost.
11. The network shown uses a Distance Vector Routing algorithm to determine the shortest paths between routers.



(i) Explain how router P initially constructs its distance vector based on its directly connected neighbours.
(ii) Show how router P updates its distance vector after receiving routing information from its neighbouring routers.
(iii) Using the Bellman-Ford equation**,** determine the shortest path and minimum cost from router P to each of the other routers. Show the necessary calculations
(iv) Prepare the final routing table of router P**,** showing the destination, next hop, minimum cost, and shortest path**.**